

This breakthrough in timing tech is unlocking a major opportunity

"You never actually own a Patek Philippe. You merely look after it for the next generation."

Those 16 words created one of the most successful advertising campaigns in history, and tens of billions of dollars in value for Swiss watchmaker Patek Philippe.



Source: Patek Philippe

In this month's issue

The next great disruption in timekeeping

The invisible tech that keeps the world in sync

Silicon is the new quartz killer

SITM's path to profits

America's fastest-growing stocks

This is my favorite ad of all time. As a dad, it gets me emotional thinking about passing down heirlooms to our eldest son, Ri. With that mindset, \$100,000 for a wristwatch almost seems... cheap.

The pitch to my wife would go, “Honey, for something that will last multiple generations, it’s a steal.”

Kidding aside, Swiss watchmakers like Patek and Rolex are some of the most successful and valuable brands of all time. And they’re still 100% privately owned, which adds to their allure.

But on Christmas Day, 1969, their dominance was shattered.

Japanese company Seiko unveiled the world’s first quartz wristwatch. Powered by a battery and a tiny, vibrating sliver of quartz crystal, it was 100X more accurate than Switzerland’s finest mechanical timepieces. And soon, it would cost roughly one-quarter of an equivalent Swiss watch.

Swiss watchmakers’ businesses were built on the perceived value of complex, handcrafted mechanical movements. How could cheap, battery-powered electronics beat more than a century of accumulated craft?

A ground-up rock made it possible.

A precisely cut sliver of quartz crystal vibrates at a precise frequency when an electrical voltage is applied. This vibration provides a steady, reliable clock signal.

For half a century, the quartz crystal has been the undisputed king of timing. And it’s quietly become the heartbeat of our digital economy.

From the iPhone in your pocket to the Wi-Fi router in your home to the data centers that power ChatGPT, they all run on a foundation of these precise timing signals. The average car contains over 100 of these tiny clocks, monitoring everything from engine performance to safety systems.

Now, we’re on the cusp of the next great disruption in timekeeping.

Think back to the last time you held a camera that used a roll of film. You had a limited number of shots and had to take the film to be developed. Then digital cameras arrived. Chemical film was replaced by a silicon computer chip.

This tech upgrade made photography instant, infinitely editable, and cheap enough to be built into every phone.

Timing technology is about to go through the same transformation. Now, a vibrating piece of rock is being replaced by a computer chip.

Today, investors are fixated on the big powerful chips and computing systems that power artificial intelligence (AI) data centers and robotaxis. They’re missing the *nervous system* that synchronizes the brain. Processing power is no longer the bottleneck. It’s the precision of their internal clocks.

AI data centers: Think of ChatGPT as a single “brain” distributed across an entire warehouse of tens of thousands of those processors. Any timing error in how these chips communicate with each other causes downtime. When a billion-dollar GPU cluster sits idle for even a few microseconds, it’s burning cash.

Robotics: A modern “lights-out” factory relies on fleets of robots working in perfect harmony. For them to operate safely at high speed, there must be zero delay.

Self-driving cars: Robotaxis are supercomputers on wheels. The car’s ability to see the world depends on syncing data from its cameras, LiDAR sensors, and radar with near-perfect precision. Timing cannot be off, even by a few milliseconds.

In short, quartz crystal can’t do the job anymore. The whole timing industry needs an upgrade, and that’s our big opportunity.

Investing in cool, little-known innovations is the best part of my job. I remember when I worked a “regular” office job, I’d spend all my time studying new technologies. It was my escape. Now, I get to do it for a living, thanks to you and thousands of other RiskHedge members.

I’ll hand you off to Chris to tell you all about this month’s pick...

The invisible tech that keeps the world in sync

Imagine conducting a symphony orchestra...

You keep the rhythm, so every musician plays their parts at precisely the right moments.

Without this precision timing, audiences couldn’t experience the brilliance of innovative composers like Bach and Beethoven.

Now imagine that orchestra has billions of musicians, all playing different instruments at speeds millions of times faster than humans can perceive.

That’s essentially what precision timing technology does for electronics—it conducts the digital symphonies being played inside these devices so we can experience the brilliance of the innovators who created them.

Every electronic device you can think of—from smartphones to self-driving cars to satellites—marches to an invisible beat. That beat is provided by timing technology—tiny “clocks” that keep the “heartbeat” of electronic devices with super-fast ticking signals.

These “ticks” ensure billions of digital operations happen at precisely the right time. They coordinate every action, from processing data to sending wireless signals to refreshing display screens.

We humans rely on hearts for a pulse and brains for intelligence. Electronics rely on timing technology for a pulse and processors for intelligence. There's no intelligence without a pulse.

For example, timing technology syncs the processors in your smartphone with memory, cameras, screens, and networks. When it fails, apps crash, calls drop, photos blur, and videos glitch.

Think about when you're on an online video call. Your device is converting your voice and video into digital data and sending it over the internet. The device on the other end is receiving those data packets and converting them back to sound and images.

If the timing of these data packets is even slightly off, you experience jitter, lag, distortion, or complete call failure.

The same principle applies throughout the digital world.

In AI data centers, for example, computing equipment, networking gear, storage systems, and management and orchestration tools must communicate and coordinate seamlessly at extremely high speeds. This ensures they can handle workloads like AI model training, inference, simulation, and data processing.

If their clocks drift apart by even small fractions of a second, data gets corrupted and operations fail.

Timing tech doesn't just keep things orderly. It also affects performance. A faster, more precise clock allows processors to do more calculations per second and ensures data is transferred without errors.

In short, timing technology is the unsung hero inside our electronic devices. It operates silently behind the scenes to provide the heartbeat all digital systems march to. And it enables amazing things like global communications, autonomous vehicles, and reliable space missions.

The problem is that multiple trends are conspiring to raise the bar for timing accuracy, reliability, and resiliency beyond the capabilities of legacy timing technology.

First, connectivity is becoming ubiquitous. From 5G networks to Wi-Fi to interconnected smart devices, countless more devices are constantly talking to each other.

Second, data speeds are increasing exponentially. Think of streaming ultra-high-def video, real-time cloud gaming, or the gigabit-per-second rates in modern fiber and wireless networks.

Both trends demand extremely accurate and reliable timing tech. When vast numbers of devices exchange huge amounts of data at ever-faster rates, even the tiniest timing errors can cause things to fall apart.

To make matters more challenging, the density of our electronics is increasing too. We're packing more and more circuits into smaller and smaller areas to improve the functionality and footprints of our devices. This creates "noisier" electrical environments with lots of interference and heat.

In these densely packed circuit boards, timing tech must be extremely resilient—able to resist interference from nearby components and endure rapid temperature changes.

As Stephen said earlier, "quartz crystal has been the undisputed king of timing" for decades.

During the mid-1900s, quartz became the gold standard for electronic timing. It enabled the first digital watches, made radio and TV broadcasting possible, and became the timing source in virtually every computer and communication system.

By the 1970s, quartz crystals were the tiny "clocks" ticking inside early microprocessors, leading to the rise of personal computers and the internet.

It's no exaggeration to say that quartz timing enabled the internet age. Every network router and PC relied on quartz timing to keep the data flowing.

Today—as we push into areas like AI supercomputing, 5G/6G communications, and autonomous vehicles and robots—precision timing is becoming even more critical.

5G networks, for example, require cell towers to stay synchronized within a few hundred nanoseconds (billionths of a second). That's about 10X more precise than 4G networks, because data moves so fast that even tiny timing offsets can cause problems.

Or think about autonomous vehicles. They might have dozens of sensors (cameras, LiDAR, radar, ultrasonic sensors) all collecting data.

To combine this sensor data and make split-second decisions, the vehicle's systems must accurately timestamp and synchronize all the information. A timing error of a few milliseconds could mean the vehicle's picture of the world is fuzzy or outdated, which could spell disaster.

As I alluded to above, the problem is that quartz can't keep up with these technological advancements.

Some of quartz's important limitations that are becoming problematic in cutting-edge applications include:

- **Size:** Engineers can't make quartz timing products small enough for an increasing number of today's advanced applications because the crystal needs a certain physical volume to resonate stably.

As we continue to miniaturize electronics, every millimeter matters more and more. If you have a sensor that's the size of a grain of rice, a millimeter-sized quartz crystal might be too big for it.

- **Fragility:** Quartz crystals are brittle. If you drop a device or subject it to too much vibration, a quartz timing chip can crack, or its frequency can wobble. That's far from ideal for the real-world stresses our electronic devices must endure.
- **Temperature sensitivity:** Quartz-based timing has a hard time with extreme temperatures or fast temperature changes. Cold or heat can make the crystal vibrate slightly slower or faster, throwing off its stability and accuracy. Rapid temperature changes can throw off its stability in similar ways.
- **Manufacturing constraints:** Quartz timing products are made using specialized crystal fabrication, cutting, and tuning processes. Each specific frequency might require a different piece of quartz cut in a particular way. This makes it rather slow and costly to introduce new frequencies or scale up production.
- **Power consumption:** While the quartz timing components themselves don't necessarily use a lot of power, their lack of stability when subjected to harsh real-world environments indirectly affects power use.

If its timing signal is not very stable, for example, an electronic device might not be able to enter low-power sleep modes for long because of the need to continuously re-synchronize or adjust for drift.

Put simply, the quartz-dominated timing tech market is in desperate need of disruption.

That's where a company called **SiTime Corp. (SITM)** comes in.

Silicon is the new quartz killer

History shows that when silicon semiconductor technology enters a field, it outperforms and replaces the legacy solution.

We saw this when silicon transistors and microchips replaced vacuum tubes, allowing engineers to build much smaller and more powerful computers...

We saw it when digital cameras with flash memory replaced film...

And we saw it when magnetic hard drives gave way to solid-state drives.

In each case, turning to silicon chips led to big gains in efficiency, size, and cost.

Now, the same thing is happening with timing. The industry is moving from quartz to silicon-based timing, and it's all thanks to SiTime.

Bosch engineer Markus Lutz and Stanford researcher Dr. Aaron Partridge founded SiTime in 2005 to make timing devices with the same silicon semiconductor technology used to make computer chips.

This idea had been around for a few decades, but no one was able to make it work until SiTime overcame the technical hurdles with patented innovations.

These timing devices work like microscopic tuning forks carved from silicon.

Like quartz crystals, they vibrate at a steady, precise frequency when you apply an electric current—like a very reliable, super-fast pendulum. But they're manufactured using the same processes that create computer chips.

The advantages of SiTime's silicon-based timing over quartz are profound:

- **Size:** SiTime can make much smaller timing devices, up to 1,000X smaller and lighter than quartz. This miniaturization—smaller than a grain of sand in some cases—means the ability to put precise clocks in places they couldn't go before, enabling new applications and form factors.

It's game-changing for mobile, wearable, Internet of Things, and other space-constrained environments like automotive systems.

- **Resiliency:** Unlike quartz, SiTime's silicon timing devices are capable of withstanding extreme shocks, vibrations, and temperatures while keeping accurate time—necessary for use in industries like aerospace, defense, and automotive.
- **Power consumption:** SiTime's silicon timing solutions consume much less power, extending battery life in portable devices and reducing energy costs in data centers.
- **Programmability:** SiTime's silicon timing devices are programmable. While quartz crystals are fixed at the frequency determined by their physical cut, silicon devices can be electronically configured to any frequency within their range.

This enables rapid customization and reduces the weeks or months of lead time required for custom quartz crystals to just hours. And it means these devices can be integrated into electronic systems in ways quartz can't.

- **Manufacturability:** Because SiTime's silicon timing devices are manufactured in standard semiconductor fabs, they benefit from the huge scale and efficiency of the chip industry.

SiTime's partners can manufacture them by the millions with consistent high quality at low cost. In contrast, quartz manufacturing is more specialized with riskier supply chains.

Bottom line: SiTime's silicon timing devices beat quartz in almost every category that matters.

The company designs a whole "toolbox" of silicon timekeeper chips, which are manufactured by its partners. Some are tiny, self-contained "ticking" devices. Others are like music producers that copy, distribute, filter, and clean those "ticks" so every chip in a smartphone, autonomous vehicle, or AI data center server agrees on the exact same beat.

SiTime dominates the silicon-based timing space with a market share north of 80%.

There are a few competitors, the most notable of which is **Microchip Technology (MCHP)**. But its silicon-based timing solutions lag SiTime's from a performance perspective. Plus, its timing business is just one piece of a much larger company.

For SiTime, timing is the sole focus—which means it devotes all its research and development to pushing timing tech forward and remaining ahead of the competition. So SiTime enjoys something close to a de facto monopoly on pure-play silicon-timing solutions.

Despite such a dominant market share, there's still tons of room for SiTime to grow because the shift from quartz-based to silicon-based timing is just getting started. Today, silicon-based timing only accounts for about 5% of the total timing tech market.

Meanwhile, SiTime has already shipped a few billion products to a customer base that reads like a who's who of technology leadership: **Apple (AAPL)**, **Google (GOOGL)**, **Microsoft (MSFT)**, **Dell (DELL)**, **HP (HPQ)**, Samsung, and countless others rely on its timing tech.

Apple's recently integrated SiTime's precision timing chips into its iPhone 16 series and new in-house modem.

I can't think of a stronger endorsement than the world's biggest consumer electronics company relying on SiTime's timing tech for its products.

SiTime's reach goes far beyond consumer electronics, of course. The company's become indispensable to the advancement of every tech trend that I can think of.

Whether we're talking about AI and data centers, 5G/6G communications, autonomous robots and vehicles, or space exploration, all the electronic devices and systems enabling these trends need precision timing. And SiTime provides the best precision-timing technology.

That means SiTime isn't just one of the most important companies in the timing market... It's one of the most important companies *in the entire world* because of its fundamental role in enabling every other technology to function.

And as the digital world becomes increasingly complex and demanding, SiTime's role will only become more critical.

SiTime's path to profits

The company's total revenue grew 83% year over year to \$60.3 million in the first quarter. And its communications, enterprise, and data center (or CED) business tripled revenue over the same period thanks to the strength of AI.

I expect AI to continue to be a big growth driver for SiTime.

AI is arguably the most demanding application for precision-timing technology. Thousands of processors must work together to analyze massive datasets, with each processor handling a small part of the overall computation.

These calculations must be perfectly synchronized because the results from one processor become the inputs for others. If timing is off by mere microseconds (millionths of a second), the entire process can fail or produce incorrect results.

SiTime's timing solutions enable this synchronization with devices that deliver 3X the accuracy of alternatives while occupying 75% less space—allowing AI systems to process information faster and more accurately than ever before.

This impact extends beyond individual AI clusters to entire data centers. As I mentioned earlier, AI infrastructure requires precisely synchronized networks of servers, storage systems, and networking equipment.

With millions of operations taking place every second, timing errors measured in nanoseconds (billionths of a second) can cause data corruption, performance degradation, or system failure.

Only SiTime's timing products provide the stability and accuracy needed for these applications while requiring significantly less power and space than traditional solutions.

I think SiTime can grow revenue at a compound annual rate of at least 50% for the next few years. Using that forecasted growth rate, I calculate a GARP Quotient of 0.44. That's firmly in the "Great" tier we talked about in [The Growth Stock Anomaly](#) special report. It indicates the stock is attractively priced for growth.

Meanwhile, the company was close to reaching free cash flow breakeven over the past year. It generated \$36.5 million in cash from operations.

After we deduct \$50 million in CapEx, we get free cash flow of negative \$13.5 million. I expect SiTime to reach free cash flow breakeven over the coming year.

Action to take: Buy SiTime Corp. (SITM) at current market prices. Set a 75% hard stop.

America's fastest-growing stocks

In the table below, we share 25 of the fastest-growing stocks in America that fit our *Disruption_X* size criteria. As a reminder, the stocks we're targeting have a market cap between \$200 million and \$20 billion.

Future recommendations may come from this list. It also gives us great insight into what's growing fast and—just as important—what's not. A stock riding on pure hype can't make this list because it requires real, bona fide revenue growth.

Note that these stocks trade on American exchanges. They aren't necessarily American companies.

We also include their raw [GARP Quotient](#) based on their most recent quarterly revenue to give an idea of how fairly valued they are.

Observations this month:

- Electric vehicle (EV) makers continue to stack the leaderboard: **Hyllion Holdings Corp. (HYLN)** maintains its spot as the second fastest-growing stock, followed by **Lotus Technology (LOT)** and China's **ZEEKR Intelligent Technology Holding Ltd. (ZK)**.

The EV space is crowded; specialized businesses will excel here. Lotus and ZEEKR have carved out niches for their luxury vehicles with advanced technology. Hyllion doesn't matter here because it was only able to grow so much in percentage terms due to its tiny revenue base. But it looks like the more mature the EV market gets, the more important differentiation becomes.

- **Sphere Entertainment Co. (SPHR)** is looking to convert big-ticket entertainment to sustainable revenue growth. The company operates the Sphere in Las Vegas—a 360 degree entertainment space that hosts everything from Backstreet Boys concerts to Hewlett-Packard's corporate events.

Yet despite the diverse entertainment lineup and robust revenue growth in the past year, Sphere's revenue took a hit in the most recently reported quarter. It could be a one-off, but the company's first quarter tends to be quite strong... so it could be a sign of more weakness to come.

- **Credo Technology Group Holding Ltd. (CRDO)**, one of our AI and data center infrastructure plays, continues to blow away Wall Street's revenue estimates thanks to strength in the AI space and customer diversification.

Remember, Credo's Active Electrical Cables (AECs) provide the high-speed, ultra-reliable, energy-efficient connections data centers need to get the most out of their AI infrastructure.

The company is guiding for 85% revenue growth in fiscal 2026 (which started on May 4, 2025) as the AI market expands and as Credo ramps up revenue from two new hyperscaler customers: **Meta Platforms (META)** and **Oracle (ORCL)**.

Based on what we see happening, we think that 85% revenue growth guidance for this year will prove conservative.

A few of the fast-growing stocks that have our attention:

AST SpaceMobile (ASTS)—Space-based cellular broadband. SpaceMobile is building out a space data service to rival Starlink. Anyone will be able to connect to satellite-based cell data through a regular, unmodified smartphone.

Partnerships will prove vital here. SpaceMobile already has agreements in place with **AT&T (T)**, **Verizon Communications (VZ)**, **Vodafone Group PLC (VOD)**, **Bell Canada Enterprises (BCE)**, and a roster of telecoms across Latin America, Africa, and Asia. The US Space Force is interested in the strategic potential of these “non-terrestrial networks” for secure defense and government communications.

SpaceMobile just started generating revenue this past year and it’s still burning a lot of cash—about \$360 million in the last year alone—so it still has a long way to go. But it ended last quarter with \$875 million in cash and raised another \$100 million since then, so it’s well-capitalized and could be a big winner in the long run if management executes.

Satellogic (SATL)—Satellite-as-a-Service provider. Sometimes you want to send a satellite into orbit to get heaps of data for your business... without going through the hassle of actually designing, prototyping, launching, and maintaining a satellite. That’s where Satellogic comes in.

The company offers its satellites through a “Constellation-as-a-Service” model, which gives any company or government entity its own, in-house satellite capabilities. We think there’s long-term potential here, but the revenue base is still quite small, with a current annual run rate of just \$13.6 million based on the \$3.4 million reported last quarter. So it’s still early days for Satellogic.

Nebius Group NV (NBIS)—AI infrastructure. Nebius continues the theme of offering hardware capabilities through a subscription service model. The company manages large clusters of graphics processing units (GPUs)—the processing power backbone of AI.

Setting up a data center is extremely capital-intensive. Nebius offers the same processing power that companies would get from having their own on-site GPU cluster, but with the convenience and cost optimization of the cloud. Revenue jumped 385% year over year to \$55.3 million in Nebius’s most recently reported quarter.

Meanwhile, based on the company’s \$161 million in revenue over the past year and \$12 billion market cap, you might think the company is overvalued. But expected growth is so strong, we’d say it’s definitely worth a deeper dive at this point.

Fastest-growing stocks in America							
Rank	Ticker	Name	Description	Market Cap (\$M)	Stock Price	Revenue Growth	GARP Quotient
1	IMMR	Immersion Corporation	Haptic software	\$279	\$8.63	3539%	0.00
2	HYLN	Hyllion Holdings Corp.	Electric truck parts	\$412	\$2.37	807%	26.08
3	ASTS	AST SpaceMobile Inc.	Space broadband	\$6,021	\$20.78	827%	253.43
4	SPHR	Sphere Entertainment Co.	Las Vegas tourism	\$1,482	\$41.39	480%	0.27
5	NBIS	Nebius Group N.V.	AI infrastructure	\$7,199	\$36.04	488%	6.67
6	ICG	Intchains Group Ltd.	Blockchain	\$260	\$4.33	385%	0.13
7	HRB	H&R Block Inc.	Tax services	\$7,562	\$55.18	4%	22.38
8	B	Barnes Group Inc.	Aerospace manufacturing	\$2,412	\$47.31	388%	0.05
9	QBTS	D-Wave Quantum Inc.	Quantum technology	\$1,637	\$5.27	122%	22.43
10	SATL	Satellogic Inc.	Satellites	\$278	\$2.90	-9%	<0.00
11	ADSE	ADS-TEC Energy PLC	Battery systems	\$804	\$15.61	268%	2.44
12	RITR	Reitar Logtech Ltd.	Logistics tech	\$221	\$3.56	187%	1.19
13	IRS	IRSA Inversiones y Representaciones Sociedad Anónima	Real estate	\$1,256	\$14.71	134%	0.00
14	EH	EHang Ltd.	Chinese drones	\$978	\$15.55	168%	5.58
15	ODD	Oddity Tech Ltd.	AI beauty tech	\$2,563	\$44.87	114%	2.10
16	CRDO	Credo Technology Group Ltd.	Data infrastructure	\$13,383	\$80.02	126%	15.57
17	ZK	ZEEKR Intelligent Technology Ltd.	Chinese EV	\$6,311	\$25.49	43%	0.17
18	MAX	MediaAlpha Inc.	Ad tech platform	\$743	\$11.14	149%	0.47
19	LOT	Lotus Technology Inc. ADR	Luxury EV maker	\$2,350	\$3.47	-8%	<0.00
20	CLSK	CleanSpark Inc.	Bitcoin miner	\$3,473	\$11.87	89%	5.36
21	AMTM	Amentum Holdings Inc.	Defense tech contractor	\$5,545	\$22.79	121%	0.33
22	INOD	Innodata Inc.	AI data services	\$1,067	\$36.78	114%	4.00
23	ACTG	Acacia Research Corporation	Patent licensing firm	\$427	\$4.39	65%	1.32
24	TKO	TKO Group Inc.	Martial arts entertainment	\$11,547	\$142.20	72%	3.15

Fastest-growing stocks in America							
Rank	Ticker	Name	Description	Market Cap (\$M)	Stock Price	Revenue Growth	GARP Quotient
25	IONQ	IonQ Inc.	Quantum technology	\$8,416	\$38.89	70%	397.70

For the most up-to-date guidance on our current positions, [click here](#) to view the portfolio online.

Chris Wood Stephen McBride

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